Infrared cylindrical cloak in nanosphere dispersed liquid crystal metamaterial

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We present a design of an infrared cylindrical cloak using nanosphere dispersed nematic liquid crystal (NLC) metamaterial following the approach of Smith's group [Science **314**, 977 (2006)]. Cloaking conditions require spatial distribution of liquid crystal birefringence with constant extraordinary index of refraction and radially dependent ordinary index of refraction. An approximate analytical formula for the latter is derived. Finite element (FE) simulations confirm the cloaking effect. Owing to the tunable birefringence of the liquid crystal component, such cloaking material offers the interesting possibilities of real-time control of invisibility. The possibility of experimental realization is briefly discussed. © 2012 Optical Society of America

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Novel metamaterials [1–16] for which the spatial distribution of optical parameters can be specifically tailor-made have proven to be a viable route to realize various linear, nonlinear, or tunable [4–8] optical properties and processes. Recently, considerable attentions have been focused on the possibility of cloaking and rendering objects invisible. Using the form-invariance of Maxwell equations under coordinate transformation one can shape the flow of light so that the target object appears to be invisible. To this end the permeability and permittivity tensors must satisfy certain conditions which are often derived by coordinate transformation [9,10]. Two main approaches to invisibility design have been developed. Firstly, invisibility carpet allowing one to hide an object on a reflective surface was realized for the infrared [11,12] and visible light [13]. Secondly, cylindrical cloaks operating only for one particular polarization of light were realized in the microwave range using split-ring resonators [1]. Recently, the cloak concept was generalized to space–time cloak [15].

The aim of this Letter is to design an infrared cylindrical cloak using nanosphere dispersed liquid crystal (NDLC) metamaterial [16,17], guided by the approaches of Smith's [1] and Shalaev's [2] groups.

An important factor in practical cloak design is loss associated with the imaginary part of the complex refractive index [18]. Previous calculations of the LC metamaterial show that loss is not prohibitively large in the region of index value around unity [19]. The loss can be further minimized by scanning the multidimensional space of design parameters. This would require massive computation efforts beyond the scope of this Letter. Loss is therefore ignored in this proof-of-concept study for the purpose of illustrating the possibility of reconfigurable cloaking with LC metamaterial.

Consider a cylindrical NLC cell with inner and outer radii a and b, filled with NDLC metamaterial (see below), with planar azimuthal alignment of NLC molecules in $x-y$ plane [Fig. 1(a)]. This order can be imposed by anchoring forces on transparent boundary surfaces of LC cell. The incident light with transverse magnetic (TM) polarization ($\vec{E}$ in $x-y$ plane and $\vec{B}$ along the axis of the cylinder) propagates in $x-y$ plane, see Fig. 1(b) and impinges as an extraordinary wave onto the NLC host. General requirements for tensor components of permittivity and permeability in TM geometry [1] state that only three tensor components enter into Maxwell equations and, correspondingly, only $(\mu'_{eff})_z$, $(\varepsilon'_{\mathrm{eff}})_r$, and $(\varepsilon'_{\mathrm{eff}})_\Theta$ must be taken into account for the cloak design. Indices z, r, Θ refer to cylindrical coordinates and primes denote real parts. Moreover, instead of controlling separately those three parameters it suffices to ensure proper values of their products $(\mu'_{\mathrm{eff}})_z(\varepsilon'_{\mathrm{eff}})_\theta$ and $(\mu'_{\mathrm{eff}})_z(\varepsilon'_{\mathrm{eff}})_r$.

Cloaking conditions for TM polarization read [1]:

$$(\mu'_{\mathrm{eff}})_z(\varepsilon'_{\mathrm{eff}})_r = \left(\frac{b}{b-a}\right)^2\left(\frac{r-a}{r}\right)^2, \tag{1}$$

$$(\mu'_{\mathrm{eff}})_z(\varepsilon'_{\mathrm{eff}})_\Theta = \left(\frac{b}{b-a}\right)^2. \tag{2}$$

The NDLC metamaterial [16,17] comprises the host nematic liquid crystal containing uniformly distributed nonmagnetic spheres with core radius r_1, made of polaritonic material and semiconductor shell (Drude material) with thickness d. The effective permittivity and permeability of NDLC were calculated in [16]:

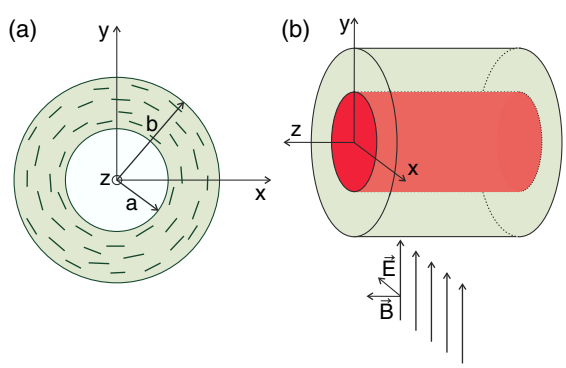

Fig. 1. (Color online) (a) Planar cylindrical NLC cell and (b) geometry of the system.

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$$\varepsilon_{\text{eff}} = \varepsilon_3\left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1}\right), \qquad \mu_{\text{eff}} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}, \quad (3)$$

where $k_3 = \sqrt{\varepsilon_3}\,2\pi/\lambda$, λ denotes the free–space wavelength, ε_3-permittivity of the NLC host, N-number density of the spheres, and a_1, b_1-scattering coefficients. Equation (3) is valid in long-wavelength limit ($\lambda/(r_1 + d) \gg 1$), for low values of filling factor $f = 4\pi(r_1 + d)^3 N/3$.

The maps of real parts $\varepsilon'_{\text{eff}}$, μ'_{eff} as functions of ε_3 and λ calculated for $r_1 = 123.5$ nm, $d = 19.5$ nm and $f = 0.06$ are shown in Fig. 2.

The tensor components of $\varepsilon'_{\text{eff}}$, μ'_{eff} are calculated using the corresponding tensor components of ε_3. In cylindrical coordinates the tensor ε_3 corresponding to planar azimuthal alignment of NLC molecules in $x - y$ plane [Fig. 1(a)] is diagonal with components $\varepsilon_r = \varepsilon_z = \varepsilon_\perp = n_o^2$, $\varepsilon_\Theta = \varepsilon_\parallel = n_e^2$. Real parts of radial and azimuthal components of effective parameters were calculated from Eq. (3): $(\varepsilon'_{\text{eff}})_r$ was calculated using $\varepsilon_3 = \varepsilon_r$ and $(\varepsilon'_{\text{eff}})_\Theta$ — using $\varepsilon_3 = \varepsilon_\Theta$.

The ultimate goal of the design is to find the spatial dependencies of NLC host parameters n_o, n_e inside the cloak domain which guarantee that cloaking relations [Eqs. (1) and (2)] hold. Below, we present an exemplary cloak design: it can be easily repeated for different sets of parameters.

Cloaking condition (1) implies that the product $(\mu'_{\text{eff}})_z(\varepsilon'_{\text{eff}})_r$ vanishes on the inner wall of the cloaking shell. To meet this requirement we put $\varepsilon'_{\text{eff}}(a) = 0$. On the other hand, $\varepsilon'_{\text{eff}}$ depends on ε_3 and λ, see Eq. (3) and Fig. 2(a). We chose $\varepsilon'_{\text{eff}}$ to vanish for $\varepsilon_3 = 2$; this value is typical for NLC matrices [17]. This yields the operating wavelength $\lambda_0 = 2.93$ μm of the cloak. Figure 3(a) shows the plots of $\varepsilon'_{\text{eff}}(\varepsilon_3)$ and $\mu'_{\text{eff}}(\varepsilon_3)$ for this wavelength. We find that the effective permeability is independent on ε_3 in the interval of values relevant for this design. Thus, we put $(\mu'_{\text{eff}})_z \equiv \mu'_{\text{eff}} \simeq 1.16$, which removes the most challenging issue of the design (similarly as in [2]) and simplifies the cloaking conditions Eq. (1) and Eq. (2) which require now that $(\varepsilon'_{\text{eff}})_\Theta$ be constant inside the cloak and $(\varepsilon'_{\text{eff}})_r$ varies with r according to (1). The former one requires, for planar azimuthal NLC geometry, that $\varepsilon_\parallel = n_e^2$ be constant. We take $n_e^2 = 4$ which is acceptable for NLC [17]. Using this value in Eq. (3) as ε_3 we find $(\varepsilon'_{\text{eff}})_\Theta \simeq 3.10$ for $\lambda_0 = 2.93$ μm. Finally, we fix the inner radius of the NLC shell ("invisibility area") to be $a = 20$ μm. Then, Eq. (2) yields the outer radius of the shell $b = 42.35$ μm. To complete the design we have to satisfy the condition (1). to this end we note that, since $(\varepsilon'_{\text{eff}})_r$ is calculated from Eq. (3) using $\varepsilon_3 = \varepsilon_\perp = n_o^2$, we have to

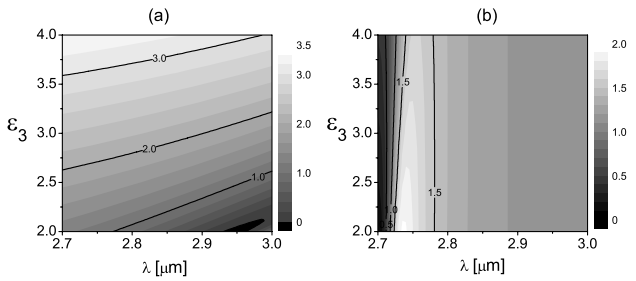

Fig. 2. (a) Maps of $\varepsilon'_{eff}(\varepsilon_3, \lambda)$ and (b) $\mu'_{\text{eff}}(\varepsilon_3, \lambda)$.

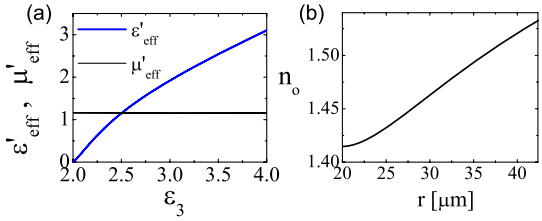

Fig. 3. (Color online) (a) Plots of real parts $\varepsilon'_{\text{eff}}$, μ'_{eff} of effective parameters (3) for $\lambda = \lambda_0 = 2.93$ μm. (b) Dependence of ordinary refractive index n_o on radial coordinate r.

ensure such a dependence of n_o on radial coordinate r that the product $(\mu'_{\text{eff}})_z(\mu'_{\text{eff}})_r(r)$ is equal to the right-hand-side of Eq. (1). This in general numerical task allows in our case an approximate analytical solution. According to Eq. (1) this product takes values from interval (0,1) and can be well approximated by a linear dependence on $\varepsilon_3 = \varepsilon_\perp$: $(\mu'_{\text{eff}})_z(\mu'_{\text{eff}})_r = 2.884\,\varepsilon_\perp - 5.772$ in the interval $2 \le \varepsilon_\perp \le 2.5$, see Fig. 3(a). Solving Eq. (1) for $n_o = \sqrt{\varepsilon_\perp}$ yields $n_o(r) = \sqrt{a_1 - a_2/r + 10a_2/r^2}$, with $a_1 = 3.25$, $a_2 = 49.8$, see Fig. 3(b).

To evaluate the performance of this particular cloak design we made field-mapping simulations using finite element package (COMSOL Multiphysics). The object hidden inside the cloak is an ideal metallic cylinder with radius a. The outgoing waves were absorbed at all the boundaries in the simulation domain. Figure 4 shows the z-component of magnetic field distribution around the metallic cylinder surrounded by the NDLC shell. Without the NDLC shell [Fig. 4(a)] a shadow is cast behind the cylinder while in the presence of the cloak [Fig. 4(b)] the wave fronts flow around the cloaked region with remarkably small perturbation. Power reflection due to the impedance mismatch at the outer cloak boundary calculated along the lines of [2] is about 10%. A deviation from azimuthal orientation of director axis caused, e.g., by turning on external electric or optical fields [4–8,20–23] can reduce the cloaking effect and allows a dynamic control of the invisibility effect. For example, the cloaking effect vanishes for a radial distribution of director axis (obtained, e.g., by applying voltage to boundary surfaces of LC cell), see Fig. 4(c); local ε_{eff}, μ_{eff} for this distribution are calculated from Eq. (3).

In the experimental realization the radial distribution of ordinary index $n_o(r)$ can be approximated by a stepwise distribution and implemented using cylindrical NLC layers separated by IR transparent walls, with constant values of n_o and n_e in each layer. The maximization of cloaking effects requires numerical optimization procedures [14,24] and will be presented elsewhere.

We have designed a cylindrical infrared cloak for TM illumination using aligned nano dispersed nematic LC. FE simulations confirm the theoretical design and the possibility of a dynamic control of the invisibility effect. The reconfigurable cloak proposed in this Letter can find important applications in complex environment where the parameters of the cloak should be carefully designed according to the background material [25]. The active elements enable the real-time control of the cloak parameters, thereby greatly easing the difficulty in the design.

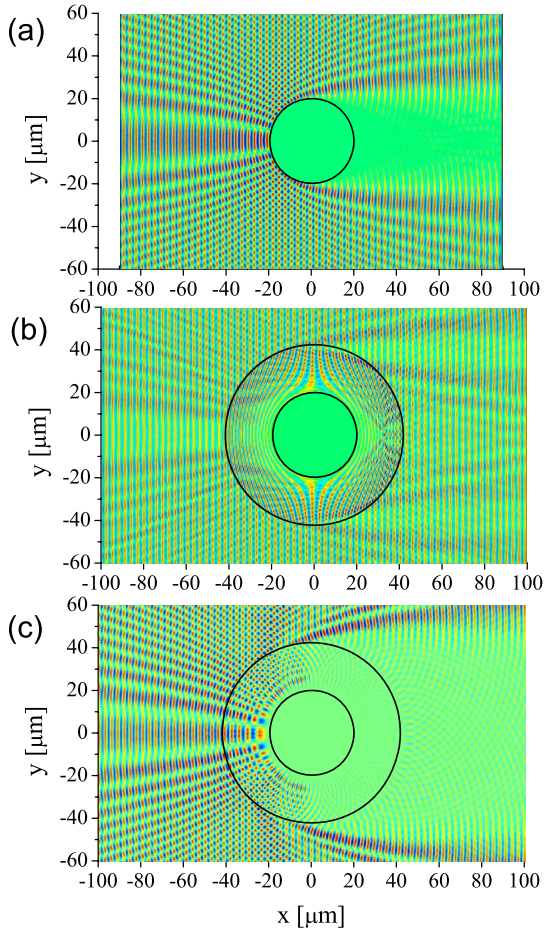

Fig. 4. (Color online) (a) Finite element simulations of the magnetic field mapping around the cloaked object with TM illumination: the object is surrounded by vacuum without the cloak, (b) the object is inside the NDLC shell cloak, (c) object inside the NDLC with radial director axis distribution-cloaking effect turned off.

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